

EduVision_DV

Higher Education Performance Dashboard

Tableau TWB / Dashboard Implementation Documentation

Purpose: Detailed technical documentation of the current Tableau workbook implementation, dashboard composition, visual encodings, KPI/measure usage, observed QA points, and the relationship between the workbook structure and the EduVision_DV requirements.

Area	Current implementation
Platform	Tableau Desktop
Primary data source	university_final_dataset
Core dashboards	University Overview; Research Analytics; Student Analytics; Country Comparison
Project role	Dashboard development / prototype-to-production transition

This document is designed to accompany the Tableau workbook in GitHub so that another developer, mentor, evaluator, or contributor can understand the TWB structure and analytical intent instead of seeing only the binary workbook artifact.

1. Workbook Architecture

The workbook is organized as a multi-dashboard analytical application. The dashboard layer controls layout and interaction, while worksheets control the underlying visual encodings, dimensions, measures and aggregations.

Logical structure

EduVision_DV ■ ■■■ Data Source ■ ■■■ university_final_dataset ■ ■■■ Analytical Worksheets ■ ■■■
Overview - Top Universities ■ ■■■ Overview - Ranking Trend ■ ■■■ Overview - Performance Scatter ■ ■■■
Research - Productivity by University ■ ■■■ Research - Impact by Region ■ ■■■ Student -
Faculty Ratio ■ ■■■ Student - Intl % by Country ■ ■■■ Country - Comparison ■ ■■■ Dashboards ■ ■■■
University Overview ■■■ Research Analytics ■■■ Student Analytics ■■■ Country Comparison

Dashboard	Visible analytical components
University Overview	Top Universities bar; Ranking Trend line; Performance Scatter
Research Analytics	Research Productivity scatter; Research Impact regional comparison
Student Analytics	Faculty Ratio comparison; International Student % map
Country Comparison	Country comparison radar/spider extension

Repository interpretation: GitHub stores the workbook artifact; Tableau Desktop is used to inspect and operate the TWB/TWBX.

2. University Overview — Implementation

The captured dashboard combines ranking, temporal movement and institution-level performance relationship. These views answer different analytical questions while sharing the same university dataset.

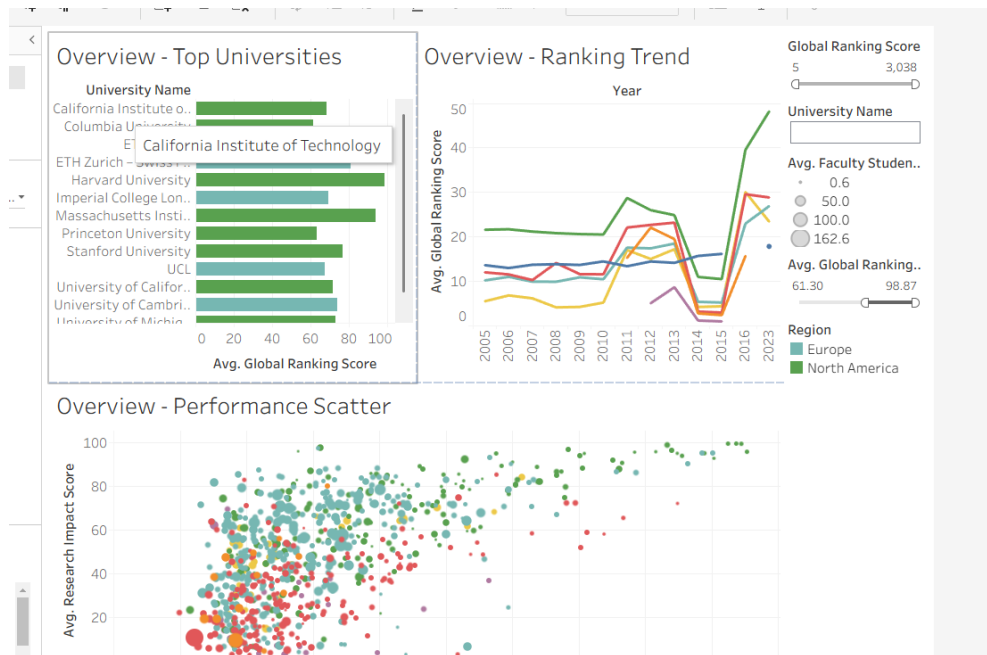


Figure 1 — Cropped Tableau capture of the University Overview dashboard.

2.1 Top Universities

- **Dimension:** University Name.
- **Measure:** AVG(Global Ranking Score).
- **Mark type:** horizontal bars.
- **Encoding:** bar length represents average ranking score; color distinguishes regions in the captured view.
- **Interactive controls visible:** University Name search and Global Ranking Score range.
- **QA:** long university names are crowded/truncated; production should provide sufficient row height or rely on full-name tooltips.

2.2 Ranking Trend

- **X:** Year.
- **Y:** AVG(Global Ranking Score).
- **Color/detail:** Region and/or university selection depending on worksheet configuration.
- **Purpose:** show movement of the ranking metric across years.
- **Visible controls:** Year range and university selection/search context.

2.3 Performance Scatter

- **X:** AVG(Academic Reputation Score).
- **Y:** AVG(Research Impact Score).
- **Size:** AVG(Faculty Student Ratio) in the captured implementation.
- **Color:** Region.

- **Detail:** University Name.

- **Interpretation:** each mark represents an institution-level observation after worksheet aggregation/context is applied.

QA observation: the scatter contains many marks. Keep valid data, but consider filter context, mark sizing/transparency, or a Top-N/detail interaction to improve readability.

3. Research Analytics — Implementation

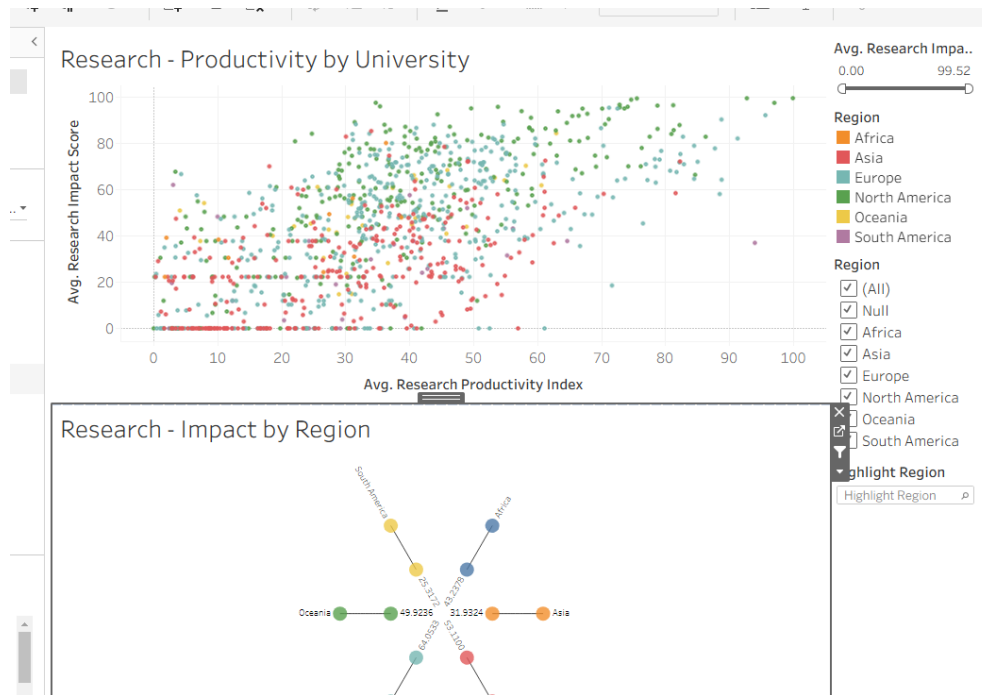


Figure 2 — Cropped Tableau capture of the Research Analytics dashboard.

3.1 Research Productivity by University

- **X:** AVG(Research Productivity Index).
- **Y:** AVG(Research Impact Score).
- **Color:** Region.
- **Mark type:** circles.
- **Visible control:** Avg. Research Impact range.
- **Role:** compare productivity and impact as a two-dimensional research-performance view.

3.2 Research Impact by Region

- The captured implementation uses a specialized visual extension for regional research impact.
- Region is encoded by color and values are based on Avg. Research Impact Score.
- The Region checklist provides an interactive inclusion/exclusion mechanism.
- The screenshot also shows a Region highlight control for focused comparison.

3.3 Research dashboard design implications

- Keep research productivity and impact measures clearly distinguished.
- Use consistent Region colors across research worksheets.
- Keep regional selectors synchronized where cross-filtering is intended.
- Document and test any third-party extension used by the workbook.

4. Student Analytics — Implementation

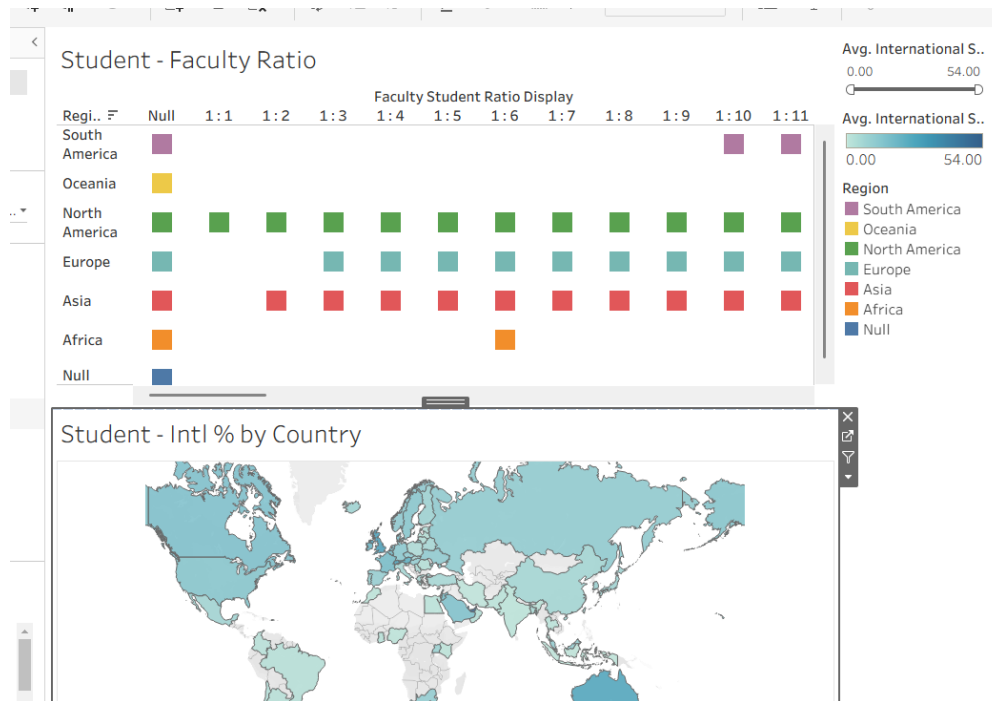


Figure 3 — Cropped Tableau capture of the Student Analytics dashboard.

4.1 Student — Faculty Ratio

- The worksheet compares Faculty Student Ratio across Region.
- The captured view exposes Faculty Student Ratio Display values and Region on rows.
- Region is color encoded.
- The captured worksheet contains a **Null** region category; review this explicitly rather than silently discarding it.

4.2 Student — International Student % by Country

- **Geography:** Country.
- **Measure:** AVG(International Student Pct).
- **Mark type:** filled geographic map.
- **Detail:** Country and University Name are visible in the Data Guide context.
- **Purpose:** show geographic variation in international student participation.

QA observation: Tableau reports **1 unknown** map mark. Resolve or document the unresolved country before final delivery.

5. Country Comparison — Implementation

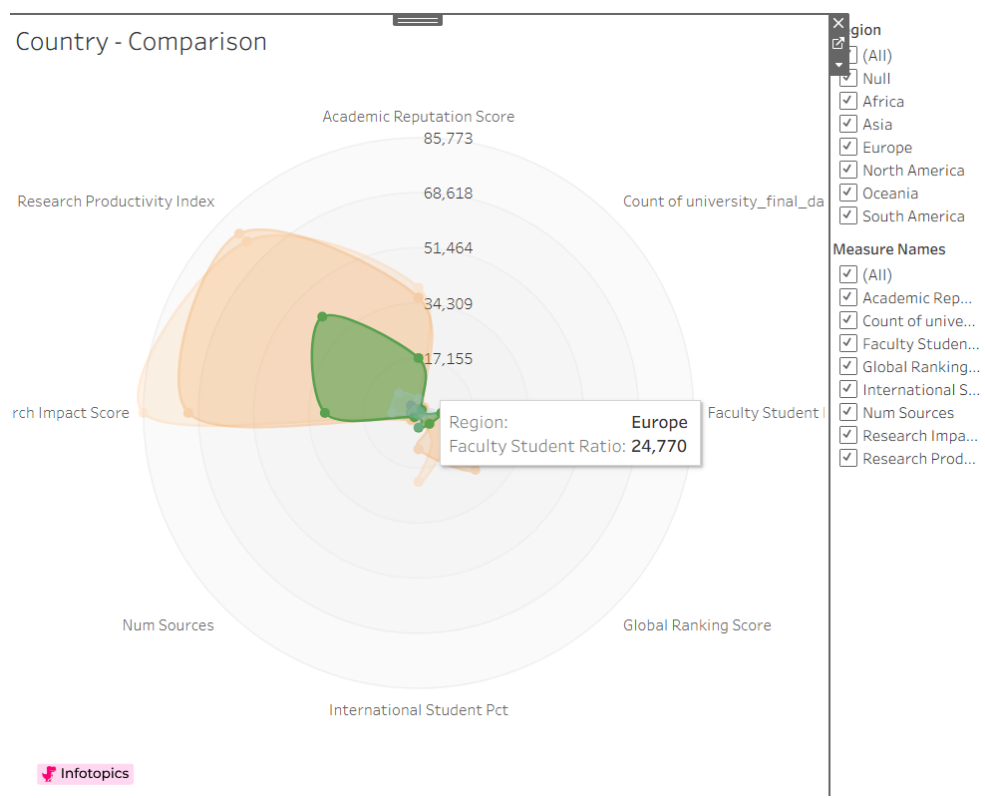


Figure 4 — Cropped Tableau capture of the Country Comparison visualization.

5.1 Radar / Spider implementation

- The current capture uses a Radar/Spider visualization extension rather than Tableau's native Bar/Line/Scatter marks.
- The visual uses Region as the comparison grouping in the displayed tooltip and color/shape encoding.
- Radial axes are generated from Measure Names, with multiple measures available in the current configuration.
- The visible Measure Names filter currently contains more measures than the intended country-comparison subset.

5.2 Required analytical measure set

Intended measure	Role in Country Comparison
Global Ranking Score	Country-level ranking comparison
Research Impact Score	Country-level research impact comparison
Academic Reputation Score	Country-level academic reputation comparison
International Student Pct	Country-level international student participation

Important refinement: the screenshot shows additional measures including Research Productivity Index, Num Sources, Faculty Student Ratio and a count measure. If the instruction book specifies four comparison measures, filter the Radar worksheet to those four.

6. KPI and Measure Engineering

Each visualization depends on a consistent aggregation strategy. These rules should remain explicit when the workbook is refined or handed to another developer.

Metric / field	Typical Tableau treatment	Use in workbook
Global Ranking Score	AVG / ranking comparison	Overview; Country Comparison
Academic Reputation Score	AVG	Overview scatter; Country Comparison
Research Impact Score	AVG	Overview; Research; Country Comparison
Research Productivity Index	AVG	Research analysis
International Student Pct	AVG	Student map; Country Comparison
Faculty Student Ratio	AVG / numeric aggregation	Student ratio; scatter size
Faculty Student Ratio Display	Text/display field	Formatted ratio presentation
University Name	Dimension / Detail	Institution-level marks
Country	Geographic dimension	Map; comparison
Region	Dimension / Color / Filter	Segmentation
Year	Discrete time dimension	Trend analysis

Aggregation discipline

- Use numeric Faculty Student Ratio for aggregation and sorting; do not aggregate the formatted display string.
- Use AVG for score-based comparisons when the dashboard requirement is an average score.
- Do not turn genuine NULLs into zero unless the business definition explicitly requires zero.
- Validate the level of detail before adding a measure: university-year, university, country-year and region are not interchangeable grains.

7. Dashboard Engineering and Interaction Layer

Layer	Responsibility
Data layer	Data source, field types, geographic roles, NULL handling, calculated fields
Worksheet layer	Rows/Columns, Marks card, aggregation, filters, color/size/detail
Dashboard layer	Containers, layout, navigation and dashboard actions

Recommended interaction map

- **Global filters:** Year, Region, Country and Subject where available.
- **Filter action:** region selections affect related views when the specification requires shared context.
- **Highlight action:** university selection/highlight can carry across connected views.
- **Parameter action:** metric selection can switch a map's encoding without changing the population.
- **Set action:** 2–5 country or university comparison can use Tableau sets.
- **Navigation:** dashboard buttons/navigation objects provide access to Overview, Research, Student and Country pages.

Action documentation rule: each action should have a documented source sheet, target sheet(s), trigger and expected effect. This prevents filters from unintentionally affecting unrelated analyses.

8. Visual QA Findings from the Supplied Captures

These are direct observations from the four supplied Tableau screenshots and should be treated as a pre-delivery checklist.

Area	Observed condition	Recommended QA action
University Overview	Long university names are crowded/truncated.	Increase row height, widen sheet, or use full-name tooltips.
University Overview	Performance scatter contains many marks.	Improve filtering, mark sizing/opacity, or provide Top-N/detail i
Research Analytics	Region checklist includes Null.	Review source mapping and decide whether NULL is valid/unknown
Student Analytics	Map displays 1 unknown.	Identify and resolve the unresolved country or document it.
Student Analytics	Faculty Ratio has a Null row.	Review Region null handling and source completeness.
Country Comparison	Radar Measure Names contains extra measures.	Restrict to the required comparison set if specified.
Country Comparison	Specialized Radar extension is used.	Document extension dependency and test portability.
General	Controls/legends consume dashboard space.	Keep decision-relevant controls visible and move technical controls

Important: these are QA observations, not claims that the underlying data is wrong. Inspect source definitions and requirements before modifying data or calculations.

9. How to Understand the TWB from GitHub

GitHub is the repository layer; Tableau Desktop is the execution and inspection layer. A reviewer should use the workbook and documentation together.

Recommended repository structure

```
Higher_Education_Dashboard_Project/ ■ ■■■ Milestone_1_Data_Collection_and_Preparation/ ■■■  
Milestone_2_KPI_Engineering_and_Dashboard_Planning/ ■ ■■■ Module_3_Education_KPI_Engineering/ ■ ■  
■■■ university_final_dataset.xlsx ■ ■ ■■ generate_education_kpis.py ■ ■■■  
Module_4_Dashboard_Planning_and_Prototyping/ ■ ■■■ EduVision_DV_Dashboard_Storyboard.pptx ■ ■■■  
eduvision_prototype.twbx ■■■ Milestone_3_Dashboard_Development/ ■■■  
Milestone_4_Testing_and_Delivery/
```

Reviewer workflow

- Read the instruction book/storyboard.
- Inspect the dataset fields and grain.
- Download the TWBX from GitHub.
- Open it in Tableau Desktop.
- Review Data Source → Worksheets → Dashboards.
- Map dashboard components to worksheets using this document.
- Test filters/actions and inspect calculations when a number appears unexpected.

Why GitHub may show a TWBX as a file rather than a Tableau dashboard: GitHub stores the artifact but does not execute Tableau Desktop. Download the TWBX and open it in Tableau, or publish it to an appropriate Tableau web environment for browser viewing.

10. Milestone 3 / Final Delivery Checklist

Check	Status to verify
Data source	Workbook opens and all data paths resolve.
University Overview	Required worksheets are visible and readable.
Research Analytics	Research visuals and filters behave correctly.
Student Analytics	Map has no unexplained unknowns; ratio view handles NULLs appropriately.
Country Comparison	Only required comparison measures are exposed.
KPIs	Aggregations and formats match business definitions.
Filters / Actions	Global filters, highlight/filter/navigation/parameter/set actions are tested.
Tooltips	Full names and relevant measures are available.
Visual design	Titles, labels, legends, spacing and hierarchy are presentation-ready.
Performance	Dashboards open and respond acceptably.
Packaging	Final TWBX contains required local data/resources.
GitHub	Correct workbook and documentation are stored under the correct milestone.

Final note

This document is based on the supplied Tableau captures and the EduVision_DV project structure. It explains the visible workbook behavior and identifies QA items without claiming that a screenshot exposes every hidden calculated field or action. Those items should be inspected directly in Tableau Desktop during final QA.